

Laboratory Equipment Booking and Maintenance System

Database implementation, relational design, SQL, reporting, triggers, and testing.

12

connected relational tables

MySQL implements the complete database layer for laboratory operations.

DATABASE

lab_equipment

Central database for laboratory equipment booking and maintenance data.

DBMS

MySQL 8.0.46

Relational implementation with SQL-defined structure, data, and behavior.

IMPLEMENTED SCOPE

Complete layer

12 tables, keys, constraints, sample data, queries, JOINS, 4 views, 2 triggers, and database testing.

The database separates laboratory operations into 12 connected tables.

Core Laboratory (lab_id) Equipment_Category (category_id) Equipment (equipment_id; category_id, lab_id)	Booking Users (user_id) Time_Slots (slot_id) Bookings (booking_id; user_id, equipment_id, slot_id) Usage_Logs (log_id; booking_id)	Maintenance Consumables (consumable_id; lab_id) Calibrations (calibration_id; equipment_id) Breakdowns (breakdown_id; equipment_id, user_id) Maintenance_Jobs (job_id; breakdown_id, vendor_id, equipment_id) Vendors (vendor_id)
--	---	--

Primary-key and foreign-key relationships connect the laboratory ER model.

CORE ENTITIES

Laboratory

1:N Equipment
1:N Consumables

Equipment_Category

1:N Equipment

BOOKING FLOW

Users

1:N Bookings and Breakdowns

Time_Slots

1:N Bookings

Bookings

1:N Usage_Logs

MAINTENANCE LINKS

Equipment

1:N Bookings, Calibrations,
Breakdowns, and
Maintenance_Jobs

Breakdowns

1:N Maintenance_Jobs

Vendors

1:N Maintenance_Jobs

The Chen-style ER model was converted into the relational schema through primary-key and foreign-key links.

The schema applies normalization through 3NF while preserving data integrity.

01

1NF

Atomic values and no repeating groups.

USER

user_id → name, email, role

02

2NF

Non-key attributes depend on the complete key.

EQUIPMENT

equipment_id →
equipment_name, category_id,
lab_id, status

03

3NF

Entity information is separated into appropriate relations and transitive dependencies are avoided where applicable.

BOOKING

booking_id → user_id,
equipment_id, slot_id,
booking_date, status

Integrity rules include PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, DEFAULT, and CHECK.

DDL, DML, and DQL define, populate, and validate the database.

SQL VOCABULARY

DDL · DML · DQL

CREATE TABLE · ALTER
TABLE · INSERT · SELECT ·
CREATE VIEW · CREATE
TRIGGER

DDL: CREATE TABLE, ALTER TABLE, CREATE VIEW, CREATE TRIGGER

DML: INSERT

DQL: SELECT

Representative table definition:

```
CREATE TABLE Equipment (  
    equipment_id INT PRIMARY KEY,  
    equipment_name VARCHAR(100) NOT NULL,  
    category_id INT NOT NULL,  
    lab_id INT NOT NULL,  
    status VARCHAR(20) NOT NULL DEFAULT 'Active',  
    FOREIGN KEY (category_id) REFERENCES Equipment_Category(category_id),  
    FOREIGN KEY (lab_id) REFERENCES Laboratory(lab_id)  
);
```

SELECT queries and JOINS turn normalized rows into operational answers.

QUERY COVERAGE

Active equipment, booking reports, most-used equipment, low-stock consumables, overdue calibration, open breakdowns, booking counts, never-booked equipment, approved-booking users, above-average bookings, breakdown reports, and maintenance reports.

JOINS combine records through foreign-key relationships.

```
SELECT b.booking_id, u.name AS user_name,  
       e.equipment_name, l.lab_name,  
       b.booking_date, ts.start_time,  
       ts.end_time, b.status  
FROM Bookings b  
JOIN Users u ON b.user_id = u.user_id  
JOIN Equipment e ON b.equipment_id = e.equipment_id  
JOIN Laboratory l ON e.lab_id = l.lab_id  
JOIN Time_Slots ts ON b.slot_id = ts.slot_id;
```

VIEWS

Four views package recurring JOINs and filters into reusable reports.

VIEW 01

Equipment_Overview

Equipment details with related category and laboratory context.

VIEW 02

Booking_Report

Booking, user, equipment, lab, and time-slot reporting.

VIEW 03

Low_Stock_Report

Consumables requiring stock attention.

VIEW 04

Calibration_Due_Report

Equipment with calibration work due.

Views provide consistent reporting surfaces without duplicating the underlying relational data.

Triggers reject invalid booking actions before insertion.

`check_equipment_active`

Active equipment only

Blocks bookings for non-Active equipment.
Equipment 109 with status Broken returns “Only active equipment can be booked”.

Test outcome: invalid booking rejected.

`prevent_booking_overlap`

No overlapping approvals

Blocks overlapping Approved bookings for the same equipment and date. Overlap returns “Booking overlaps with an existing approved booking”.

Test outcome: conflicting booking rejected.

Valid booking actions proceed; invalid actions are stopped before INSERT.

The implemented database is populated, queryable, report-ready, and validated.

Tables and records	Count
Laboratory · Equipment_Category · Equipment	5 · 7 · 10
Users · Time_Slots · Bookings	10 · 8 · 12
Usage_Logs · Consumables · Calibrations	6 · 8 · 8
Breakdowns · Maintenance_Jobs · Vendors	5 · 5 · 2

FINAL RESULT

Database layer complete

12 relational tables support laboratory operations. DDL, DML, DQL, JOINS, 4 views, and 2 triggers were implemented.

The stated trigger tests rejected invalid equipment and overlapping approved bookings with the expected error messages.